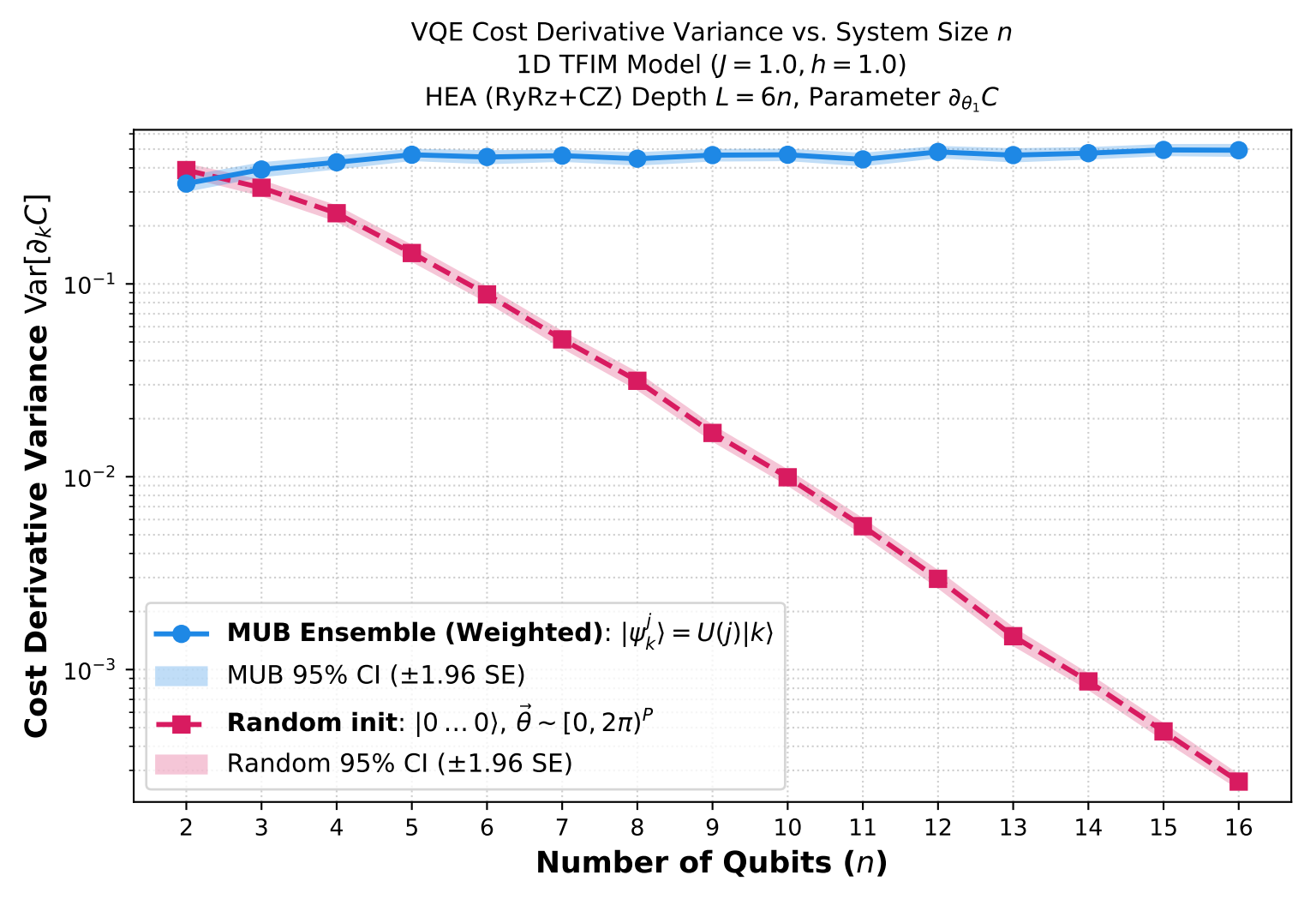
VQE Cost Derivative Variance vs. System Size n
1D TFIM Model ($J = 1.0, h = 1.0$)
HEA (RyRz+CZ) Depth $L = 6n$, Parameter $\partial_{\theta_1} C$
Cost Derivative Variance $\mathrm{Var}[\partial_k C]$
10^{-1}
10^{-2}
10^{-3}
MUB Ensemble (Weighted): $|\psi_k^j\rangle = U(j)|k\rangle$
MUB 95% CI (±1.96 SE)
Random init: $|0 \ldots 0\rangle$, $\vec{\theta} \sim [0, 2\pi)^P$
Random 95% CI (±1.96 SE)
2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
Number of Qubits (n)